

Notes on Quantum Hall Effect

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1 Integer Quantum Hall Effect

This section describes the integer quantum hall effect. The phenomena is, if we apply a magnetic field perpendicular to a two dimensional material, the conductivity is quantized in integer values. We are familiar with quantities getting quantized. These are usually “fundamental” quantities. On the other hand, the conductivity is a statistical, averaged quantity of a material. The quantization of this quantity begs an explanation. This section will attempt to understand the integer quantum hall effect.

P.S. This is just an elaboration of David Tong’s awesome notes on the same subject. I wrote this to make matters clear to myself!

1.1 The Classical Hall Effect

Take a two dimensional conductor. Let us take the plane of the conductor to be the $x - y$ plane. We then apply an electric field along the x -direction, so that there is a current I_x along the x -direction. Now, we turn on a magnetic field B along the z -direction, perpendicular to the plane of the conductor. The moving charges will experience a force due to the magnetic field along $\hat{x} \times \hat{z} = -\hat{y}$ direction. This induces a current along the $-y$ -direction. Hence, positive charges will accumulate at $-\hat{y}$ edge of the material. This makes an electric field in the $\hat{y}$ direction, which works against the magnetic field. At steady state, there will be a voltage across the y -direction, and not net current flow along y . This is what we expect. Now, how do we describe the system quantitatively? Let us look at one electron, classically. Under an electromagnetic field, the electron experiences a force given by the Lorentz force,

$$\mathbf{F} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (1.1)$$

Also, when the electron moves through the material, it collides with other electrons or impurities, or whatever that comes in its way. We model this as a friction term,

$$\mathbf{F}_{\text{fric}} = -\frac{m\mathbf{v}}{\tau} \quad (1.2)$$

Here, τ is called the *scattering time*, the average time in between collisions. The inverse relationship of friction and scattering time is intuitively clear. The larger the scattering time is, the fewer the impedance. Now, we plug all of this into Newton’s law,

$$m \frac{d\mathbf{v}}{dt} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - \frac{m\mathbf{v}}{\tau} \quad (1.3)$$

This is called the *Drude model*. We would like to understand the steady state. In the steady state, there is no net motion for the electron. Hence we can set the left hand side of the above equation to be zero. This gives,

$$\mathbf{v} + \frac{e\tau}{m}\mathbf{v} \times \mathbf{B} = -\frac{e\tau}{m}\mathbf{E} \quad (1.4)$$

We know that the current density is,

$$\mathbf{J} = n(-e)\mathbf{v} \quad (1.5)$$

Using this, we get,

$$\mathbf{J} - \frac{e\tau}{m}\mathbf{J} \times \mathbf{B} = \frac{ne^2\tau}{m}\mathbf{E} \quad (1.6)$$

The cross product is done easily,

$$\mathbf{J} \times \mathbf{B} = B(\hat{x}J_y - \hat{y}J_x) \quad (1.7)$$

Hence,

$$\begin{pmatrix} 1 & -\omega_B\tau \\ \omega_B\tau & 1 \end{pmatrix} \mathbf{J} = \frac{ne^2\tau}{m}\mathbf{E} \quad (1.8)$$

Where $\omega_B = \frac{eB}{m}$ is the *cyclotron frequency*. The matrix is easily inverted to write,

$$\mathbf{J} = \frac{ne^2\tau/m}{1 + (\omega_B\tau)^2} \begin{pmatrix} 1 & \omega_B\tau \\ -\omega_B\tau & 1 \end{pmatrix} \mathbf{E} \equiv \sigma \mathbf{E} \quad (1.9)$$

Here σ is a conductivity tensor. We clearly see that even if we apply an electric field along the x -direction, there is a current along the y -direction due to the off diagonal elements of the conductivity tensor. This off diagonal element is called the Hall conductivity. It is convenient to actually look at the inverse, called the resistivity tensor,

$$\rho = \sigma^{-1} = \frac{1}{\sigma_{\text{DC}}} \begin{pmatrix} 1 & -\omega_B\tau \\ \omega_B\tau & 1 \end{pmatrix} \quad (1.10)$$

where $\sigma_{\text{DC}} = \frac{ne^2\tau}{m}$ is the conductivity without the magnetic field. The off diagonal term is,

$$\rho_{xy} = \frac{\omega_B\tau}{\sigma_{\text{DC}}} = \frac{B}{ne} \quad (1.11)$$

Note that this is independent of the scattering time. We also have,

$$\rho_{xx} = \frac{m}{ne^2\tau} \quad (1.12)$$

which depends on the scattering time. Therefore, here is our classical prediction for Hall effect : If we increase the magnetic field, ρ_{xx} stays constant, and ρ_{xy} increases linearly with the magnetic field. This, of course is not what is observed in a lab!

1.2 The Quantum Hall Effect

Let us first describe what is actually seen in the lab. For certain materials, as the magnetic field is increased, ρ_{xy} stays a constant for a certain range of magnetic field values! And then upon further increase after a particular value of B , ρ_{xy} takes another constant value larger than before, and stays there for a while before taking another step after a certain magnetic field strength is reached, and this continues! For the range of value of B for which ρ_{xy} stays a constant, ρ_{xx} goes to zero. And when ρ_{xy} makes a jump, ρ_{xx} sharply peaks! The plateaus of resistivity was found to obey

$$\rho_{xy} = \frac{2\pi\hbar}{e^2} \frac{1}{\nu} \quad \text{where } \nu \in \mathbb{Z} \quad (1.13)$$

The classical picture do not capture this at all. Definitely, quantum mechanics is doing something interesting. Let me try to unpack this here.

1.2.1 Landau Levels

Let us go step by step. First, let us understand the quantum mechanics of a charged spinless particle on a magnetic field. First let me explain why we can get away with spinless assumptions when we study electrons, which are obviously spinning. The reason is that when we introduce a magnetic field, it introduces an energy gap between the two spins, $\Delta E = 2\mu_B B$, where μ_B is the Bohr magneton. If we restrict ourselves to very large magnetic field values, then it takes a large amount of energy to flip the spin. So, if we restrict to low energy large magnetic field setting, the particles are effectively spinless. We assume this is the case.

The Lagrangian for a particle in a magnetic field is,

$$L = \frac{1}{2}m\dot{\mathbf{x}}^2 - e\dot{\mathbf{x}} \cdot \mathbf{A} \quad (1.14)$$

The vector potential enjoys a gauge symmetry,

$$\mathbf{A}(\mathbf{x}) \rightarrow \mathbf{A}(\mathbf{x}) + \nabla\alpha(\mathbf{x}) \quad (1.15)$$

If we make this change, the Lagrangian changes by a total time derivative of $\alpha(\mathbf{x})$, and hence the classical physics stays the same. But the fact that the vector potential, not the magnetic field is what appears in the Lagrangian is responsible for a lot of interesting physics in the quantum world. The canonical momentum is,

$$\mathbf{p} = m\dot{\mathbf{x}} - e\mathbf{A} \quad (1.16)$$

The canonical momentum is not gauge invariant. The (gauge invariant) mechanical momentum is,

$$\boldsymbol{\pi} = m\dot{\mathbf{x}} = \mathbf{p} + e\mathbf{A} \quad (1.17)$$

The Hamiltonian is,

$$H = \frac{1}{2m}(\mathbf{p} + e\mathbf{A})^2 = \frac{1}{2m}\boldsymbol{\pi}^2 \quad (1.18)$$

We see that the Hamiltonian involves only the mechanical momentum, which we already knew in the guise of the lesson “magnetic fields do no work”. First, we solve it classically to get an intuitive picture. This is simple, we just have to use Lorentz force law and Newton’s law,

$$m\frac{d\mathbf{v}}{dt} = -e\mathbf{v} \times \mathbf{B} \quad (1.19)$$

Which gives,

$$m\ddot{x} = -eB\dot{y} \quad \text{and} \quad m\ddot{y} = +eB\dot{x} \quad (1.20)$$

This is easily solved to get,

$$x(t) = X - R\sin(\omega_B t + \phi) \quad \text{and} \quad y(t) = Y + R\cos(\omega_B t + \phi) \quad (1.21)$$

The particles moves in circles with frequency ω_B of radius R fixed by initial conditions and the circle is centered at (X, Y) , also fixed by initial condition.

Now, let us write down the Poisson bracket structure, in preparation for canonical quantization. We have,

$$\{x^i, p_j\} = \delta_j^i \quad \text{and} \quad \{x^i, x^j\} = 0 = \{p_i, p_j\} \quad (1.22)$$

The mechanical momentum components do not commute,

$$\{\pi_i, \pi_j\} = \{p_i + eA_i, p_j + eA_j\} = -e\left(\frac{\partial A_j}{\partial x^i} - \frac{\partial A_i}{\partial x^j}\right) = -e\epsilon_{ijk}B_k \quad (1.23)$$

Now, we are ready to quantize. This is done by promoting $\{\cdot, \cdot\} \rightarrow \frac{1}{i\hbar}[\cdot, \cdot]$. Thus we have,

$$[x^i, p_j] = i\hbar\delta_j^i \quad (1.24)$$

Also,

$$[\pi_i, \pi_j] = -ie\hbar\epsilon_{ijk}B_k \quad (1.25)$$

In particular,

$$[\pi_x, \pi_y] = -ie\hbar B \quad (1.26)$$

This prompts us to define,

$$a = \frac{1}{\sqrt{2e\hbar B}}(\pi_x - i\pi_y) \quad \text{and} \quad a^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\pi_x + i\pi_y) \quad (1.27)$$

These operators obey the Harmonic oscillator algebra,

$$[a, a^\dagger] = 1 \quad (1.28)$$

Then, the Hamiltonian takes the form,

$$H = \hbar\omega_B \left(a^\dagger a + \frac{1}{2} \right) \quad (1.29)$$

This is the same as the harmonic oscillator, and the energy levels (called Landau levels) are,

$$E_n = \hbar\omega_B \left(n + \frac{1}{2} \right) \quad (1.30)$$

There is one thing that is different from the harmonic oscillator. Here, each energy level is degenerate, with a very high number of degeneracy. We missed this along the way we were trying to find the spectrum¹.

Landau Gauge

Let us do the calculation in another way, which will let us find the wave functions and degeneracy. Since the choice of gauge is up to us, we choose what is known as the Landau gauge,

$$\mathbf{A} = xB\hat{y} \implies \nabla \times \mathbf{A} = \mathbf{B} \quad (1.31)$$

Now, the Hamiltonian reads,

$$H = \frac{1}{2m} (p_x^2 + (p_y + eBx)^2) \quad (1.32)$$

Even though we were in the infinite plane where we had translational invariance along x and y , the choice of gauge broke the translational invariance along x . But we still have invariance along y . We make use of this fact and make a guess for the wave function,

$$\psi_k(x, y) = e^{iky} f_k(x) \quad (1.33)$$

Plugging this into the Schrodinger equation,

$$\frac{1}{2m} (p_x^2 + (\hbar k + eBx)^2) f_k(x) = E_k f_k(x) \quad (1.34)$$

¹the SHO is one dimensional while the Landau problem is in two dimensions. We surely missed one dof.

This is nothing but a shifted harmonic oscillator (not surprising anymore). We can re-write it in a bit more apparent form,

$$\left(\frac{p_x^2}{2m} + \frac{1}{2}m \left(\frac{eB}{m} \right)^2 \left(x + \frac{\hbar k}{eB} \right)^2 \right) f_k(x) = E_k f_k(x) \quad (1.35)$$

We have $\omega_B = \frac{eB}{m}$ and define a length scale $\ell_B^2 = \frac{\hbar}{eB}$. Then,

$$\left(\frac{p_x^2}{2m} + \frac{1}{2}m\omega_B^2(x + \ell_B^2 k)^2 \right) f_k(x) = E_k f_k(x) \quad (1.36)$$

This is indeed a shifted harmonic oscillator with the center at $-k\ell_B^2$, and the energy levels are precisely the Landau levels. Importantly, the energies do not depend on k . This is the degeneracy I mentioned earlier. The wave functions are,

$$\psi_{n,k}(x, y) \sim e^{iky} H_n(x + k\ell_B^2) \exp \left\{ -\frac{(x + k\ell_B^2)^2}{2\ell_B^2} \right\} \quad (1.37)$$

Now, we can see the degeneracy as well. The degeneracy is infinite because k takes continuous values. But in real life, we have a finite size material. Let us take a rectangular block of sides L_x and L_y . The y momentum k cannot take infinitely many values now, but can only take discrete values in steps of $\frac{2\pi}{L_y}$. Along x , we did not have an infinite momentum, in fact it is a localised state. But there is a restriction, which is : where the wave function is localised cannot be outside the material. That is, we must have,

$$0 \leq -k\ell_B^2 \leq L_x \implies 0 \geq k \geq -\frac{L_x}{\ell_B^2} \quad (1.38)$$

Therefore, k takes values in between the above range in steps of $\frac{2\pi}{L_y}$. This is the same for all levels n . Hence, the degeneracy in one level is,

$$\mathcal{N} = \frac{\text{total range of values of } k}{\text{steps size}} = \frac{\int_{-L_x/\ell_B^2}^0}{2\pi/L_y} = \frac{L_x L_y}{2\pi \ell_B^2} = \frac{eBA}{2\pi \hbar} \quad (1.39)$$

where $A = L_x L_y$ the area of the sample. It is customary to introduce another constant called the *quantum of flux*,

$$\Phi_0 = \frac{2\pi \hbar}{e} \quad (1.40)$$

the,

$$\mathcal{N} = \frac{AB}{\Phi_0} \quad (1.41)$$

Symmetric Gauge

There is nothing special about the Landau gauge. In fact, in certain situations, it is useful to use other gauges. Let me do the same problem once more, but this time using the symmetric gauge. This is the choice,

$$\mathbf{A} = -\frac{1}{2}yB\hat{x} + \frac{1}{2}xB\hat{y} = -\frac{1}{2}\mathbf{r} \times \mathbf{B} \quad (1.42)$$

We see that this gauge breaks translational invariance in x and y directions. But it preserves the rotational symmetry in the xy plane about the origin. Therefore, angular momentum is a good quantum number.

Now we solve this using the algebraic method. We defined ladder operators for $\boldsymbol{\pi}$ and then could immediately derive the spectrum. To get the wave functions, we have to supplement the dof that we omitted. We use the following,

$$\tilde{\boldsymbol{\pi}} = \mathbf{p} - e\mathbf{A} \quad (1.43)$$

This differs in the sign of e from the mechanical momentum. Clearly, this is not gauge invariant. If we recall,

$$[\pi_x, \pi_y] = -ie\hbar B \quad (1.44)$$

then the commutation relation for $\tilde{\boldsymbol{\pi}}$ is obtained by flipping the sign of e . Therefore,

$$[\tilde{\pi}_x, \tilde{\pi}_y] = +ie\hbar B \quad (1.45)$$

Therefore, one might think that we can define ladder operators for the tilde “momentum” in similar manner and diagonalise both $\boldsymbol{\pi}$ and $\tilde{\boldsymbol{\pi}}$ at the same time. But that would be incorrect, since these two do not commute with each other in general. We can find,

$$[\pi_x, \tilde{\pi}_x] = 2ie\hbar \frac{\partial A_x}{\partial x}; \quad [\pi_y, \tilde{\pi}_y] = 2ie\hbar \frac{\partial A_y}{\partial y}; \quad [\pi_x, \tilde{\pi}_y] = [\pi_y, \tilde{\pi}_x] = ie\hbar \left(\frac{\partial A_x}{\partial y} + \frac{\partial A_y}{\partial x} \right) \quad (1.46)$$

However, if we stick with symmetric gauge, all the above commutators vanish! Therefore, we can diagonalise these two operators simultaneously in the symmetric gauge. Hence define,

$$b = \frac{1}{\sqrt{2e\hbar B}}(\tilde{\pi}_x + i\tilde{\pi}_y) \quad \text{and} \quad b^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\tilde{\pi}_x - i\tilde{\pi}_y) \quad (1.47)$$

These look almost like the a and $a^\dagger$ definition, except we have different signs in the definition. a looks the tilded version of $b^\dagger$. The reason is the sign flip in the algebra. With this definition, we get,

$$[b, b^\dagger] = 1 \quad (1.48)$$

We then define the ground state to be annihilated by both a and b . And have a general state,

$$|n, m\rangle = \frac{(a^\dagger)^n (b^\dagger)^m}{\sqrt{n!m!}} |0, 0\rangle \quad (1.49)$$

And the energy spectrum is the Landau levels, which is independent of m . Therefore, m is the degeneracy of each level.

Let us now construct the wave functions. First, we focus on the lowest Landau level with $n = 0$. The wave functions obey,

$$a |0, m\rangle = 0 \quad (1.50)$$

We know,

$$\begin{aligned} a &= \frac{1}{\sqrt{2e\hbar B}} (\pi_x - i\pi_y) \\ &= \frac{1}{\sqrt{2e\hbar B}} (p_x + eA_x - i(p_y + eA_y)) \\ &= \frac{1}{\sqrt{2e\hbar B}} \left[\left(-i\hbar \frac{\partial}{\partial x} - \frac{eB}{2} y \right) - i \left(-i\hbar \frac{\partial}{\partial y} + \frac{eB}{2} x \right) \right] \\ &= \frac{1}{\sqrt{2e\hbar B}} \left[-i\hbar \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) - i \frac{eB}{2} (x - iy) \right] \end{aligned} \quad (1.51)$$

The above form is screaming at us to use complex coordinates. Let us define,

$$z = x - iy \quad \text{and} \quad \bar{z} = x + iy \quad (1.52)$$

correspondingly,

$$\partial = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \quad \text{and} \quad \bar{\partial} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \quad (1.53)$$

Hence,

$$a = \sqrt{\frac{\hbar}{2eB}} \left[-i2\bar{\partial} - i \frac{eB}{2\hbar} z \right] = -i\sqrt{2}\ell_B \left(\bar{\partial} + \frac{z}{4\ell_B^2} \right) \quad (1.54)$$

similarly,

$$a^\dagger = i\sqrt{2}\ell_B \left(\partial + \frac{\bar{z}}{4\ell_B^2} \right) \quad (1.55)$$

Therefore, the lowest level wave function has to take the form,

$$\psi_{n=0,m}(z, \bar{z}) \sim f(z) \exp \left\{ -\frac{|z|^2}{4\ell_B^2} \right\} \quad (1.56)$$

By playing a similar game, we can write,

$$b = -i\sqrt{2}\ell_B \left(\partial + \frac{\bar{z}}{4\ell_B^2} \right) \quad \text{and} \quad b^\dagger = i\sqrt{2}\ell_B \left(\bar{\partial} + \frac{z}{4\ell_B^2} \right) \quad (1.57)$$

Then, we discover that for $m = 0$, the $f(z)$ written above has to be a constant. Then acting upon by $b^\dagger$ increases powers of z by one. Hence,

$$\psi_{n=0,m}(z, \bar{z}) \sim \left(\frac{z}{\ell_B} \right)^m \exp \left\{ -\frac{|z|^2}{4\ell_B^2} \right\} \quad (1.58)$$

These eigen states are also simultaneous eigen states of angular momentum. The angular momentum operator is,

$$J = i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = \hbar(z\partial - \bar{z}\bar{\partial}) \quad (1.59)$$

If we act on the ground state,

$$\hbar(z\partial - \bar{z}\bar{\partial})\psi_{n=0,m}(z, \bar{z}) = \hbar \left(m - \frac{z\bar{z}}{4\ell_B^2} - \left(-\frac{\bar{z}z}{4\ell_B^2} \right) \right) \psi_{n=0,m}(z, \bar{z}) = m\hbar\psi_{n=0,m}(z, \bar{z}) \quad (1.60)$$

The angular momentum depends on which b level you are in, which means that for each state in the same energy level, the angular momentum distinguishes it.

We can find out the degeneracy in a different way now. We consider a disc shaped region of radius R . How many states of the same Landau levels lie within this disk? For Landau level zero, the state with angular momentum is peaked around a radius $r = \sqrt{4m\ell_B^2}$. This is easy to see, use $z = re^{i\theta}$ and take an r derivative,

$$0 = \frac{\partial}{\partial r} \left[\left(\frac{z}{\ell_B} \right)^m \exp \left\{ -\frac{|z|^2}{4\ell_B^2} \right\} \right] \implies r^2 = 4m\ell_B^2 \quad (1.61)$$

As we increase the angular momentum, the radius square at which the wave function is peaked increases linearly, in steps of $2\ell_B^2$. Therefore, the number of states in a given radius square R^2 is,

$$\mathcal{N} = \frac{R^2}{\text{step size}} = \frac{\pi R^2}{2\pi\ell_B^2} = \frac{eAB}{2\pi\hbar} \quad (1.62)$$

which is the same as before.

1.2.2 Turning on Electric Field

Now, let us turn on the electric field on top of the magnetic field. Remember, we want to turn on the electric field in the x direction, and then we expect a voltage across the y direction over time, which is the Hall effect. Turning on an electric field corresponds to a change in the Hamiltonian,

$$H = \frac{1}{2m}(\mathbf{p} + e\mathbf{A})^2 + eEx \quad (1.63)$$

Let us work in the Landau gauge. Then,

$$H = \frac{1}{2m}(p_x^2 + (p_y + eBx)^2) + eEx \quad (1.64)$$

The same analysis we did before works. The momentum along y is still conserved, hence make the guess,

$$\psi_k(x, y) = e^{iky} f_k(x) \quad (1.65)$$

Plugging this into the Schrodinger equation will give,

$$\left(\frac{p_x^2}{2m} + \frac{1}{2}m\omega_B^2(x + \ell_B^2 k)^2 + eEx \right) f_k(x) = E_k f_k(x) \quad (1.66)$$

This is again a shifted harmonic oscillator, we just need to complete the square,

$$\left(\frac{p_x^2}{2m} + \frac{1}{2}m\omega_B^2 \left(x + \ell_B^2 k + \frac{eE}{m\omega_B^2} \right)^2 - \frac{1}{2}m\omega_B^2 \left(\frac{e^2 E^2}{(m\omega_B^2)^2} + k\ell_B^2 eE \right) \right) f_k(x) = E_k f_k(x) \quad (1.67)$$

Therefore, the wave functions are,

$$\psi_{n,k}(x, y) \sim e^{iky} H_n \left(x + \ell_B^2 k + \frac{eE}{m\omega_B^2} \right) \exp \left\{ -\frac{\left(x + \ell_B^2 k + \frac{eE}{m\omega_B^2} \right)^2}{2\ell_B^2} \right\} \quad (1.68)$$

Now, the energy is,

$$E_{n,k} = \hbar\omega_B \left(n + \frac{1}{2} \right) - eE \left(k\ell_B^2 + \frac{eE}{2m\omega_B^2} \right) \quad (1.69)$$

With the electric field, the degeneracy is lifted. The energy now depends linearly on k . Since the energy is dependent on the y momentum, we expect that the eigen states drift in the y direction. The group velocity of the drift is,

$$v_y = \frac{1}{\hbar} \frac{\partial E_{n,k}}{\partial k} = -\frac{eE}{\hbar} \ell_B^2 = -\frac{E}{B} \quad (1.70)$$

Therefore, we can write the energy levels in a suggestive way,

$$\begin{aligned}
E_{n,k} &= \hbar\omega_B \left(n + \frac{1}{2} \right) - eE \left(k\ell_B^2 + \frac{eE}{m\omega_B^2} \right) + \frac{eE}{2} \frac{eE}{m\omega_B^2} \\
&= \hbar\omega_B \left(n + \frac{1}{2} \right) - eE \left(k\ell_B^2 + \frac{eE}{m\omega_B^2} \right) + \frac{1}{2}m \left(\frac{E}{B} \right)^2 \\
&= \hbar\omega_B \left(n + \frac{1}{2} \right) - eE \left(k\ell_B^2 + \frac{eE}{m\omega_B^2} \right) + \frac{1}{2}mv_y^2
\end{aligned} \tag{1.71}$$

The last two terms above look like the kinetic energy due to drift plus the potential energy of the charged particle being at position $-\left(k\ell_B^2 + \frac{eE}{2m\omega_B^2}\right)$. The total energy is then the center of mass energy plus excitations! Thus, if we put an electric field alongside a magnetic field to a two dimensional material, then the electrons drift not in the direction of $\mathbf{E}$, but in the direction $\mathbf{E} \times \mathbf{B}$. This is what we saw from our classical analysis as well.

1.2.3 A Partial Explanation of Hall Quantization

In this section, I will explain the value of the Hall quantization plateau in several different ways. Why the plateau exists in the first place will be explained in the next section. First, let us look at a “semi-classical” explanation. We know from experiments that the Hall resistivity takes quantized values given by,

$$\rho_{xy} = \frac{\Phi_0}{e} \frac{1}{\nu} \quad \text{and} \quad \rho_{xx} \rightarrow 0 \tag{1.72}$$

with $\nu \in \mathbb{N}$. We also learned that when we turn on magnetic field perpendicular to a two dimensional plane in which electrons move, the energy levels are quantized and are labelled by a natural number,

$$E_{\nu,k} = \hbar\omega_B \left(\nu + \frac{1}{2} \right) \tag{1.73}$$

The naive thing to do is to relate these two natural numbers. i.e, the Hall resistivity takes the quantized value given by the number ν when precisely ν Landau levels are filled. This naive guess turns out to give the correct answer. Let us see how. We derived the Hall resistivity classically from the Drude model to be,

$$\rho_{xy} = \frac{B}{ne} \tag{1.74}$$

If ν Landau levels are filled, the electron density is,

$$n = \frac{\mathcal{N}}{A} \times \nu = \frac{B}{\Phi_0} \nu \tag{1.75}$$

Plugging this in,

$$\rho_{xy} = \frac{\Phi_0}{e} \frac{1}{\nu} \quad (1.76)$$

This explains the quantization at the plateau!

If ν Landau levels are completely filled, then there is a gap $\hbar\omega_B$ to the next level. So if we are at small temperature compared to this gap, $k_B T \ll \hbar\omega_B$, then even if we apply an electric field along x , then the electrons cannot move anywhere, since there is no available spot in the level! Therefore there will be no conduction in the x direction. Hence $\rho_{xx} \rightarrow 0$.

Doing a Better Job

This simple explanation, which is actually a mix of classical and quantum results, seems to give us the correct quantization rule for the Hall resistivity. But we can do a better job, by calculating the conductivity using quantum mechanics (as opposed to using classical results). Let us think about one electron, and its current density. For an electron, the charge density is $-\frac{e}{A}$. Then, the current density is,

$$\mathbf{J}_{\text{one particle}} = -\frac{e}{A} \dot{\mathbf{x}} = -\frac{e}{mA} (\mathbf{p} - e\mathbf{A}) \quad (1.77)$$

Let us use the Landau gauge to perform further calculations, hence $\mathbf{A} = xB\hat{y}$. Apply an electric field along x , then the total current in the material is the expectation value of the current for all filled states. Let us assume that ν Landau levels are filled. Then, the total current along the x direction is,

$$J_x = -\frac{e}{mA} \sum_{n=1}^{\nu} \sum_k \langle \psi_{n,k} | p_x | \psi_{n,k} \rangle \quad (1.78)$$

Recall that in the Landau gauge with the electric field, the wave functions took the form,

$$\psi_{n,k}(x, y) \sim e^{iky} H_n \left(x + \ell_B^2 k + \frac{eE}{m\omega_B^2} \right) \exp \left\{ -\frac{\left(x + \ell_B^2 k + \frac{eE}{m\omega_B^2} \right)^2}{2\ell_B^2} \right\} \quad (1.79)$$

This is a free particle along y and harmonic oscillator along x . We know that the average momentum of the harmonic oscillator is zero. Therefore,

$$J_x = 0 \quad (1.80)$$

Now, the total current along the y direction is,

$$J_y = -\frac{e}{mA} \sum_{n=1}^{\nu} \sum_k \langle \psi_{n,k} | (p_y - eBx) | \psi_{n,k} \rangle \quad (1.81)$$

Expectation value of p_y is $\hbar k$, and expectation value of the position x is where the harmonic oscillator is centered, which is at,

$$-\ell_B^2 k - \frac{eE}{m\omega_B^2} = -\frac{\hbar k}{eB} - \frac{mE}{eB^2} \quad (1.82)$$

Thus,

$$J_y = e\nu \frac{E}{AB} \sum_k \quad (1.83)$$

The sum over ν is performed before the sum over k because the number of values of k is independent of the level². Sum over k is just,

$$\sum_k = \mathcal{N} = \frac{AB}{\Phi_0} \quad (1.84)$$

therefore,

$$J_y = \frac{e}{\Phi_0} \nu E \quad (1.85)$$

In vector notation,

$$\begin{pmatrix} J_x \\ J_y \end{pmatrix} = \begin{pmatrix} 0 & \frac{e}{\Phi_0} \nu \\ -\frac{e}{\Phi_0} \nu & 0 \end{pmatrix} \begin{pmatrix} E \\ 0 \end{pmatrix} \quad (1.86)$$

Taking the inverse, we get the resistivity,

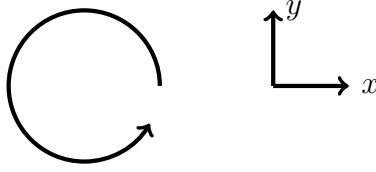
$$\rho_{xy} = \frac{\Phi_0}{e} \frac{1}{\nu} \quad \text{and} \quad \rho_{xx} = 0 \quad (1.87)$$

Which again, is the thing that we see in the labs!

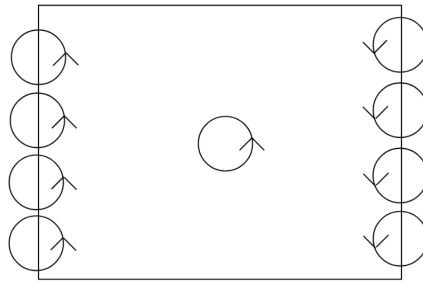
1.2.4 Edge Modes

Thus far, we were discussing materials with infinite size along x and y . Let me remedy this now. Consider a sample with finite size along x and infinite size along y . Let us switch the electric field off. To set our expectations, let us analyse the situation classically. In the bulk of the sample, electrons do the usual thing. They form orbits. If the applied magnetic field is along z , then the orbits will look like this (assuming a small current along x),

²remember, we no longer have degeneracy when the electric field is on.

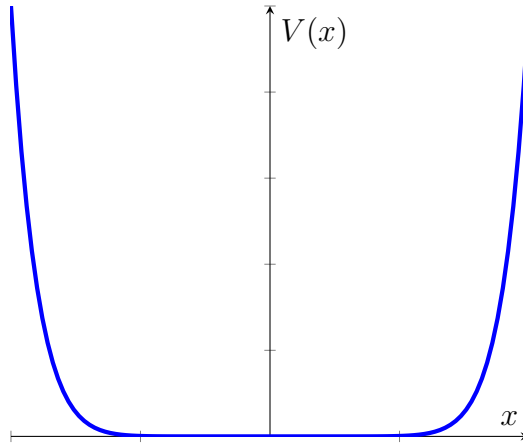


Now, at the edges along x , the electrons won't be able to complete the orbit. On the left edge say, the orbit will hit the boundary. When the electron hits the boundary, it just bounces back, and does the anti clockwise motion again until it hits the boundary again. Overall there will be a flow of electron in the y direction at the left edge, and $-y$ direction at the right edge. A figure of it will look like this,



The overall current is still zero, because the currents at the edges cancel.

Now let us describe the edge effect quantum mechanically. Let me keep the electric field off for simplicity. The edge effect can be thought of as including a potential in the Hamiltonian, which is almost like the infinite well potential,



We know the wave functions of the system when the bounding potential is absent. Now, the width of these wave functions is of the order ℓ_B . Let x_0 be the center of the wave function.

We can then Taylor expand the wave function around the wave function's center,

$$V(x) = V(x_0) + (x - x_0) \frac{\partial V(x_0)}{\partial x} + \dots \quad (1.88)$$

We neglect the constant $V(x_0)$ term, and all higher order terms starting from quadratic. This is by the assumption that over the length scale ℓ_B , the potential varies linearly³. Therefore for a wave function centered at x_0 , the effect of the bounding potential is like adding an “electric field”,

$$“E_x” = \frac{1}{e} \frac{\partial V(x_0)}{\partial x} \quad (1.89)$$

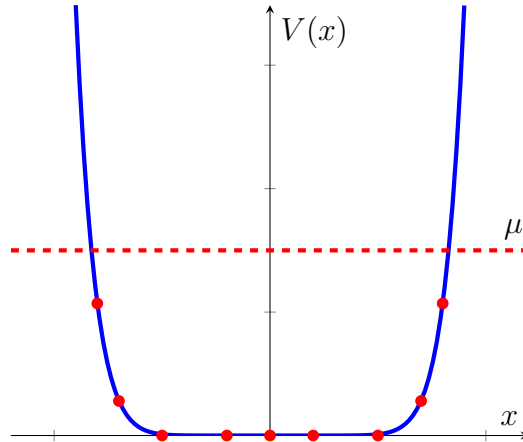
the Hamiltonian should be modified with $e“E_x”x$. We have already studied this effect. The electrons will get a drift velocity along the y direction given by,

$$v_y = -\frac{E}{B} = -\frac{1}{eB} \frac{\partial V(x_0)}{\partial x} \quad (1.90)$$

If the electron is on the right edge of the potential, then slope is positive and hence gets a drift along $-y$, and therefore overall current along $+y$. The opposite is true on the left edge. This is exactly what we saw from our classical picture.

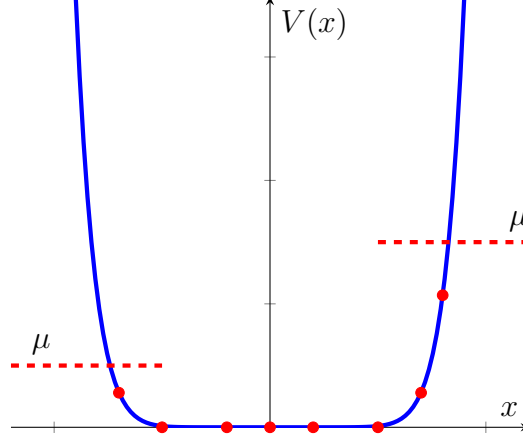
One More Stab at the Hall Plateau

Let us now assume that we have a chemical potential, and then we can fill in states up to this value. Assume that we only have one Landau level. Since where the state is localised is determined by the k value, instead of filling in states by k values, we can equally well think of filling in states at specific x position. Let us fill all the states in the Landau level up to the chemical potential, and it will look something like this,



³For any smooth function, if we look close enough, it appears linear.

Clearly, in the bulk of the sample, the filled states have nowhere to move to upon excitation hence is an insulator. But at the edges, the electrons have band left beyond μ , hence behaves as metals. It is hence clear that as long as the temperature is low enough, there is no conduction along x . Now, let us apply a potential difference between the two edges,



Then, we know that there is going to be overall current now, because of the applied potential difference⁴. The overall drift is along y direction as we know, and the current is,

$$I_y = -e \int \frac{dk}{2\pi} v_y(k) \quad (1.91)$$

Since labelling the state by k is equivalent to labeling it by x (where the wave packet is centered), change from $k \rightarrow x$ using $x = k\ell_B^2$,

$$I_y = \frac{e}{2\pi\ell_B^2} \int dx \frac{1}{eB} \frac{\partial V(x)}{\partial x} = \frac{e}{2\pi\hbar} \Delta V \quad (1.92)$$

The applied voltage (also called the Hall voltage) is,

$$V_H = e\Delta V \quad (1.93)$$

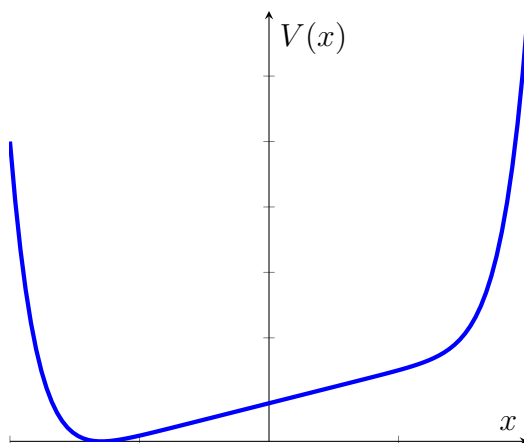
which gives us the Hall conductivity,

$$\sigma_{xy} = \frac{I_y}{V_H} = \frac{e^2}{2\pi\hbar} \quad (1.94)$$

This is the conductivity for one Landau level as expected.

⁴remember, when no field is applied, there is edge currents but the total current is zero because the edge currents at opposite edges are equal and opposite.

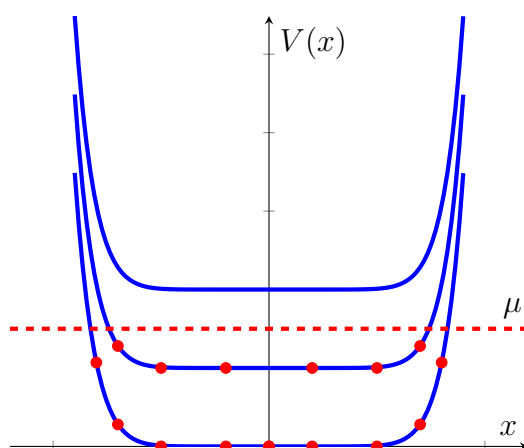
It seems like that the Hall conductivity is completely due to currents at the edge and the bulk has no role⁵. This is only true if we have zero applied electric field. If we apply an electric field, we have seen that the degeneracy is lifted and the potential looks like,



Then the drift in the bulk is also not zero because the potential is not flat precisely because of the electric field. Now when we apply a potential difference, the Hall conductivity will be coming from both the edge and bulk y currents.

Notice that it does not matter what the shape of the potential in the middle for the value of the Hall conductivity. A random shape in the middle would imply disorder, and this picture hand wavingly shows why even with disorder we will have the quantized values of the Hall conductivity.

Now if extend the story to having multiple Landau levels filled, like below,



⁵In the bulk, the drift along y was zero because the potential is flat.

As long as the chemical potential is between two Landau levels, we would have the same current, except for a multiplicative factor for how many Landau levels are filled. Therefore, we arrive back at the quantized values of the Hall conductivity for ν Landau levels filled.

1.2.5 A Full Explanation of Hall Quantization

So far, I have only explained the conductivity values when ν Landau levels are filled. But what the experiment reveals to us is that, The conductivity stays on a plateau for a range of value of the magnetic field. Recall the degeneracy density at a Landau level is,

$$n = \frac{B}{\Phi_0} \nu \quad (1.95)$$

Suppose fix the density of electrons. Then we know from the previous section that if the magnetic field reaches a value $B = \frac{n\Phi_0}{\nu}$, then ν Landau levels are completely filled and the conductivity gets the quantized value. But what if the magnetic field is not exactly this value? Naively we would expect that the xy conductivity will be different from the quantized value. Also, in this case the last Landau level will be partially filled. This means that there are accessible states along x direction and we expect a conductivity along x .

But the experiments reveals to us the opposite. The hall conductivity stays at the quantized value corresponding to ν as the magnetic field is changed from $\frac{n\Phi_0}{\nu}$ to $\frac{n\Phi_0}{\nu-1}$, before jumping to the quantized value corresponding to the integer $\nu - 1$, and the xx conductivity remain zero through out, only peaks at the transition of plateau.

An Aside : What is “Happening” in an Experiment?

If the above paragraph is confusing, this is for you. I am not going to explain what experimenters are doing, since I have no clue. But I will explain the thought experiment we did above. We fix the electron density n in a material. Then, for a very large value of the magnetic field, the degeneracy at each Landau level is really high, typically way larger than n . Therefore, all $n \times A$ electrons in the material can fit in the first Landau level, and still have spots left.

Now, we *decrease* the magnetic field. Then, then number of states available in each level decreases. Therefore, after a particular value of the field, there are not enough spots in the first Landau level for the $n \times A$ electrons. Therefore, they have to go to the next Landau level.

Disorder Explains the Plateau

Ironically, what explains the neat quantized values and the plateau of the Hall conductivity is the disorder in the system! Let me first explain how disorder explains the plateau. The disorder in a material can be thought of as a random potential $V(\mathbf{x})$. We will ask two things from the potential : First, the potential has a very small strength compared to the Landau level splitting,

$$V \ll \hbar\omega_B \quad (1.96)$$

This condition smears out the Landau level, but not too much. Let me explain. Typically, a perturbing potential will lift the degeneracy. We expect V to do the same as well. Hence each Landau level becomes a band. But since the strength of the perturbation is very small compare to the level splitting, the bands of one level do not cross the band of the next level. Therefore, we just have a smeared band centered around each Landau level.

The second demand on the potential is that it changes only in large length scales compared to the natural length ℓ_B . i.e,

$$|\nabla V| \ll \frac{\hbar\omega_B}{\ell_B} \quad (1.97)$$

Since the cyclotron orbits are of the radius of order ℓ_B , this assumption lets us approximate each cyclotron orbit to be happening in a constant V .

With these two assumptions on the potential, we can prove the following claim : The center of the cyclotron orbit (X, Y) drifts perpendicular to ∇V . Let us prove this first. Recall, we have the center of the orbits to be,

$$X = x(t) + R \sin(\omega_B t + \phi) \quad \text{and} \quad Y = y(t) - R \cos(\omega_B t + \phi) \quad (1.98)$$

This can be re-written as,

$$X = x(t) - \frac{\dot{y}}{\omega_B} = x + \frac{\pi_y}{m\omega_B} \quad \text{and} \quad Y = y + \frac{\pi_x}{m\omega_B} \quad (1.99)$$

here, $\boldsymbol{\pi}$ is the mechanical momentum. We can quantize this and then have quantum operators for the center of the orbit. In the absence of the potential, it is easy to show that $\dot{X}$ and $\dot{Y}$ are zero. But now with the potential present, we have,

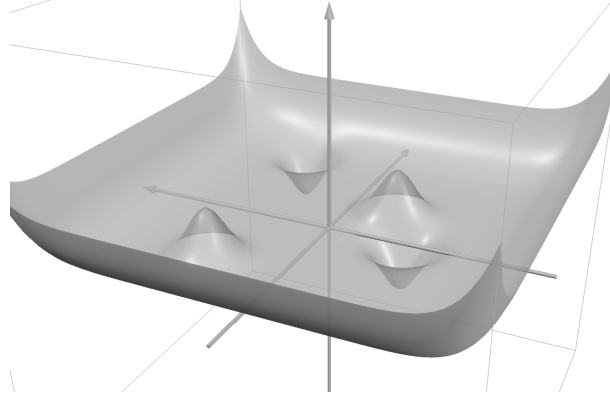
$$i\hbar\dot{X} = [X, H + V] = [X, V] = [X, Y] \frac{\partial V}{\partial Y} = i\ell_B^2 \frac{\partial V}{\partial Y} \quad (1.100)$$

and,

$$i\hbar\dot{Y} = [Y, H + V] = [Y, V] = [Y, X] \frac{\partial V}{\partial X} = -i\ell_B^2 \frac{\partial V}{\partial X} \quad (1.101)$$

Therefore, $(\dot{X}, \dot{Y})$ is in the direction $(\frac{\partial V}{\partial Y}, -\frac{\partial V}{\partial X})$, which is orthogonal to $\nabla V = (\frac{\partial V}{\partial X}, \frac{\partial V}{\partial Y})$.

What does this imply? We know that orthogonal direction to the gradient is equipotentials. Therefore, the cyclotron center drifts along equipotentials. If the potential has a positive peak, then the movement is in the anti clockwise direction, and if the peak is negative, the movement is clockwise⁶. An example random potential will look like this,



At the ends, we have peaks for the boundary of the material. In the middle, we have random troughs and peaks. We see that there are two types of equipotential lines for this random potential : one, that is localised around an extrema in the bulk, two, ones that extends from one end of the material to the other. The second kind exists mainly because of the potential at the edge. This has equipotential lines that extends throughout the material. This types of equipotential splits the states into two bins. Ones that are localised and ones that are extended. A state that is in the bulk is likely to be trapped on an equipotential line around an extrema, which are the localised states and a state which is close to the edge is extended because it drifts from one end to the other of the sample. In the band of energy at each Landau level, the ones that are at the high or low energy ends are localised, while the ones that stay near the central region of the band is extended.

It is then clear that only the extended states transport current through the sample. Therefore, as we decrease the magnetic field, first the conducting excited states are filled, which gives a certain value for the conductivity. As we decrease the magnetic field further, the localised states gets filled, but that do not change the conductivity! Therefore, the conductivity stays constant, until the magnetic field is decreased to a value so that there are not

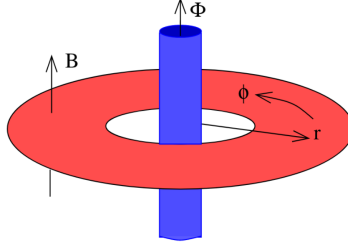
⁶Take $V = x^2 + y^2$, then gradient at $(1, 1)$ is $(2, 2)$, then the orbit takes the direction $(2, -2)$ which is anti clockwise.

enough space in the present Landau level. Then the electrons jump down to the next Landau level, and the conductivity goes to the next Plateau!

Disorder Explains the Hall Conductivity

Now, we explained how disorder leads to the plateau. But, this seems to imply that the value of the conductivity is going to be different from what we initially derived using no impurity assumptions. Since the localised states do not participate in current conduction, the conductivity must be different from what we initially found. Yet again, disorder saves the day and explains the value of the conductivity as well. Let us see how.

First, we change the setup a little bit. Instead of looking at a rectangular sample, we will now look at an annular sample. Applying a voltage around the sample will then be achieved by adding an extra magnetic flux Φ through the center on top of the applied magnetic field through the sample, as shown below,



If we increase the flux Φ from zero to Φ_0 in some time $T \gg 1/\omega_B$, then Faraday's law gives us the EMF induced around the sample to be,

$$\mathcal{E} = -\frac{\partial\Phi}{\partial t} = -\frac{\Phi_0}{T} \quad (1.102)$$

Now, let us set the disorder to zero first, and consider an electron moving in a ring (not on the annulus for now) in the presence of the flux. We know even though the electron never sees the flux, it still feels the effects quantum mechanically. This is the Aharonov Bohm effect. The reason is that outside the region where flux is zero, the vector potential is non zero and takes the form,

$$A_\phi = \frac{\Phi}{2\pi r} \quad (1.103)$$

Then, the Hamiltonian reads,

$$H = \frac{1}{2m}(p_\phi + eA_\phi) = \frac{1}{2mr^2} \left(-i\hbar \frac{\partial}{\partial \phi} + \frac{e\Phi}{2\pi} \right)^2 \quad (1.104)$$

The eigen states are,

$$\psi_n = \frac{1}{\sqrt{2\pi r}} e^{-in\phi} \quad \text{where } n \in \mathbb{Z} \quad (1.105)$$

Then the spectrum is,

$$E_n = \frac{1}{2mr^2} \left(\hbar n - \frac{e\Phi}{2\pi} \right)^2 = \frac{\hbar^2}{2mr^2} \left(\hbar n - \frac{\Phi}{\Phi_0} \right)^2 \quad (1.106)$$

We see that the flux affects the energy levels, unless the flux is an integer multiple of Φ_0 . Now, if we increase the flux from zero to Φ_0 , the n th level becomes the $(n+1)$ th level. This is known as the spectral flow.

Let us get back to the annulus, but still with zero disorder. Since the system have rotational symmetry, it is wise to choose the symmetric gauge. We saw that the wave functions in the symmetric gauge are,

$$\psi_{n=0,m}(z, \bar{z}) \sim \left(\frac{z}{\ell_B} \right)^m \exp \left\{ -\frac{|z|^2}{4\ell_B^2} \right\} = \frac{r^m}{\ell_B^m} e^{im\phi} \exp \left\{ -\frac{r^2}{4\ell_B^2} \right\} \quad (1.107)$$

Where I used $z = re^{i\phi}$. Then, the m quantum number is the angular momentum quantum number. Now, if we increase the flux from zero to Φ_0 , then we know that the state at m will flow to the state at $m+1$. This implies that a state is physically transported to a larger radius. This follows from the fact that we discussed earlier : a state with angular momentum m is localized around the radius $r = \sqrt{2m\ell_B^2}$ (assuming we choose m such that the radius lies inside the annulus). Therefore, as the spectrum flows, states are transported radially. We have,

$$\psi_m(\Phi = 0) \longrightarrow \psi_m(\Phi = \Phi_0) = \psi_{m+1}(\Phi = 0) \quad (1.108)$$

Therefore, we can think of increasing the flux as equivalent to shifting the states radially. If all the states in the Landau level was filled to start with, then this radial shifting will leave a hole at the inner radius. This is equivalent to transporting one electron from the inner radius to the outer radius. If ν Landau levels are filled, then ν electrons are transported from the inner to outer ring. Therefore, the radial current is,

$$I_r = -\frac{\nu e}{T} \quad (1.109)$$

Hence the Hall resistivity is,

$$\rho_{r\phi} = \frac{\mathcal{E}}{I_r} = \frac{\Phi_0}{e} \frac{1}{\nu} \quad (1.110)$$

Just as before.

Now, let us turn the disorder back on. In the presence of disorder, we have seen there are two types of states : extended and localised. And we know that only extended states contribute to conduction. This should mean that dialing the flux from zero to Φ_0 should not affect the localised states at all. i.e, The localised states should not undergo spectral flow. Let us see how.

Changing the flux from zero to some Φ will change the wave function by a phase. But we can simply undo this by a gauge transformation,

$$\psi \rightarrow \exp\left\{-\frac{ie\Phi\phi}{2\pi\hbar}\right\}\psi \quad (1.111)$$

But is the gauge transformation valid? Let us take the extended states first. These have non zero support for the entire range of ϕ . Therefore, we would want to ensure the single valuedness of these states. That is, as we change the angle $\phi \rightarrow \phi + 2\pi$, we should get back the same state. Therefore, if the above gauge transformation is possible, we must have,

$$\exp\left\{-\frac{ie\Phi\phi}{2\pi\hbar}\right\} = \exp\left\{-\frac{ie\Phi(\phi + 2\pi)}{2\pi\hbar}\right\} \quad (1.112)$$

This is true only if $\Phi \in \Phi_0\mathbb{Z}$. But as we increase the flux from zero to Φ_0 , this is not true. Therefore, we cannot do a gauge transformation on the extended states.

On the other hand, localised states do not have support over the full range of ϕ , and hence there is no need to ensure single valuedness as ϕ changes from 0 to 2π . For the localised states, we can gauge transform and nullify the effect of the change of flux.

Therefore, the spectrum unchanged in the presence of disorder as well if the flux is a multiple of Φ_0 . But as we change the flux from zero to Φ_0 , the localised states do not change and therefore do not undergo a spectral flow, but the extended states do get affected and undergoes spectral flow. There are at least two extended states, one at the inner ring, one at the outer ring. As the spectral flow happens, the extended states gets shifted radially outwards, so exactly one hole is left at the inner ring. This is equivalent to transporting one electron from the inner to the outer ring. If we transport one electron from the inner to outer, then we know that the conductivity takes the quantized values!

Let me summarise. We have the annular material, and a fixed magnetic field B applied through it. We fix the number density of electrons, and start with a very high value of B . Then, the degeneracy at each Landau level is high, and let us suppose that B is such that all electrons are in the first Landau level. Now, we apply an EMF along ϕ direction in the guise of a separate flux threaded through the center. As we change the flux from zero to Φ_0 , we see that there is a current along r direction, and the conductivity is the quantized $\nu = 1$ value. Now, with the flux on, we decrease the magnetic field. As the field is decreasing, the number of spots in the Landau level decreases. After some value of B , there are not enough space in the level, the electrons jumps to the next Landau level. Let us suppose that the states gets filled in the order of energy. At each level, the lowest lying energy states are localised, and as these states gets filled, the conductivity stays at the $\nu = 1$ value. As these low lying localised states are filled, we get to the extended states. When they get filled, the conductivity changes to $\nu = 2$ value. After, that, we have the localised states with higher energy, and as they get filled, the conductivity will remain at $\nu = 2$ value, and will continue to be in this value until the $\nu = 3$ extended states are reached. Phew!

1.3 Hall Effect on the Torus

We will now consider a topologically interesting two dimensional material, the torus T^2 . If the torus is imagined to be resting on the xy plane, we apply a uniform magnetic field in the z direction. Now, let us put wave functions on it. Since we have periodicity in the space, the wave functions must have some periodicity property. We demand wave functions after going around a circle agree, up to a gauge transformation. After all, we expect that the wave functions will pick up a phase as it moves through a magnetic field. Let us formulate this using the translation operator. Define,

$$T(\mathbf{d}) = \exp\left\{-\frac{i}{\hbar}\mathbf{d} \cdot \mathbf{p}\right\} = \exp\left\{-\frac{i}{\hbar}\mathbf{d} \cdot (-i\nabla + \frac{e\mathbf{A}}{\hbar})\right\} \quad (1.113)$$

It is clear that the translation is gauge dependent. Let us fix a gauge, say the Landau gauge, $A_x = 0$ and $A_y = xB$ ⁷. We then demand as the boundary conditions,

$$\psi(x, y) = T_x\psi(x, y) = \psi(x + L_x, y) \quad (1.114)$$

and,

$$\psi(x, y) = T_y\psi(x, y) = \exp\left\{-\frac{ie x L_y B}{\hbar}\right\}\psi(x, y + L_y) \quad (1.115)$$

⁷Now, x, y means the directions along the torus, rather than the plane on which it rests!

Here, T_x and T_y are translations using $\mathbf{d} = (L_x, 0)$ and $\mathbf{d} = (0, L_y)$ respectively. We have what we wished. As the wave functions are translated around the torus, they come back to the original wave function up to a phase.

But, there is still a possible issue. If we do two consecutive translations, we get results that do not agree.

$$T_x T_y \psi(x, y) = T_x \exp\left\{-\frac{ie x L_y B}{\hbar}\right\} \psi(x, y + L_y) = \exp\left\{-\frac{ie x L_y B}{\hbar}\right\} \psi(x + L_x, y + L_y) \quad (1.116)$$

and,

$$T_y T_x \psi(x, y) = T_y \psi(x + L_x, y) = \exp\left\{-\frac{ie(x + L_x)L_y B}{\hbar}\right\} \psi(x + L_x, y + L_y) \quad (1.117)$$

Therefore, we must have,

$$\exp\left\{-\frac{ie L_x L_y B}{\hbar}\right\} = 1 \implies B = \frac{2\pi\hbar n}{e L_x L_y} \quad \text{where } n \in \mathbb{Z} \quad (1.118)$$

The possible magnetic field values that we can put on the torus is restricted! Notice that if we demand the stricter boundary condition $\psi(x + L_x, y) = \psi(x, y + L_y) = \psi(x, y)$, the only magnetic field that we can put on the system is $B = 0$.

Now, we add fluxes. Remember, we used an extra flux to turn on electric field before. We are going to do the same here. But now, we have two circular directions, therefore can have two fluxes. We put both. Φ_x and Φ_y . Then the vector potential is,

$$\mathbf{A} = \left(\frac{\Phi_x}{L_x}, \frac{\Phi_y}{L_y}\right) + \mathbf{A}_{\text{old}} \quad (1.119)$$

Here, $\mathbf{A}_{\text{old}}$ is the vector potential for the applied field $\mathbf{B}$. Changing these fluxes will introduce an electric field, and which in turn will result in currents. Overall, the Hamiltonian will get a new term,

$$\Delta H = -\mathbf{J} \cdot \mathbf{A} = - \sum_{i=x,y} \frac{J_i \Phi_i}{L_i} \quad (1.120)$$

We need to evaluate the conductivity due to this perturbation. In general, this is answered through the Kubo formula, which is the quantum mechanics generalisation of Ohm's law. Let us derive that now,

Kubo Formula

Let us denote the free Hamiltonian by H_0 and its solutions,

$$H_0 |m\rangle = E_m |m\rangle \quad (1.121)$$

Also, we choose to work in temporal gauge, where we set $A_0 = 0$. Then,

$$\mathbf{E} = -\partial_t \mathbf{A} \quad (1.122)$$

Since $\mathbf{E}$ is the applied field, we choose it to be a monochromatic field,

$$\mathbf{E}(t) = \mathbf{E} e^{-i\omega t} \implies \mathbf{A} = \frac{\mathbf{E}}{i\omega} e^{-i\omega t} \quad (1.123)$$

In the end if we wish, we will take $\omega \rightarrow 0$, so that we have the conductivity for a constant applied field.

We will work with the interaction picture, therefore the states obey,

$$|\psi(t)\rangle_I = U_I(t, t_0) |\psi(t_0)\rangle_I \quad (1.124)$$

Where, U_I obey the Schrodinger equation with the evolution operator being the interaction Hamiltonian, $\Delta H(t) = e^{\frac{i}{\hbar} H_0 t} \Delta H e^{-\frac{i}{\hbar} H_0 t}$. The formal solution looks like,

$$U_I(t, t_0) = \mathcal{T} \exp \left\{ -\frac{i}{\hbar} \int_{t_0}^t dt' \Delta H(t') \right\} \quad (1.125)$$

The question we ask is this : If we prepare the system in a specific state $|\psi_0\rangle$ at time $t \rightarrow -\infty$, (this is usually taken to be the many body ground state, but it doesn't necessarily have to), then what is the expectation value of the current at some time t ? We answer it the following way,

$$\begin{aligned} \langle \mathbf{J}(t) \rangle &= \langle \psi_0(t) | \mathbf{J}(t) | \psi_0(t) \rangle \\ &= \langle \psi_0 | U_I^{-1}(t) \mathbf{J}(t) U_I(t) | \psi_0 \rangle \end{aligned} \quad (1.126)$$

where,

$$U_I(t) = \mathcal{T} \exp \left\{ -\frac{i}{\hbar} \int_{-\infty}^t dt' \Delta H(t') \right\} \quad (1.127)$$

Now, we assume linear response. That is, we assume that the applied electric field is small, and expand the evolution operator to linear order in perturbation,

$$\langle \mathbf{J}(t) \rangle = \langle \psi_0 | \left[\mathbf{J}(t) - \frac{i}{\hbar} \int_{-\infty}^t dt' [\mathbf{J}(t), \Delta H(t')] \right] | \psi_0 \rangle \quad (1.128)$$

The first term in the above expression is the current in the absence of electric field. We will take it to be zero. Then,

$$\langle \mathbf{J}(t) \rangle = \frac{i}{\hbar} \int_{-\infty}^t dt' \langle \psi_0 | [\Delta H(t'), \mathbf{J}(t)] | \psi_0 \rangle \quad (1.129)$$

Using the form of the perturbation, we have,

$$\langle J_i(t) \rangle = \frac{1}{\hbar\omega} \int_{-\infty}^t dt' \langle \psi_0 | [J_j(t'), J_i(t)] | \psi_0 \rangle E_j e^{-i\omega t'} \quad (1.130)$$

Define a variable, $t'' = t - t'$,

$$\langle J_i(t) \rangle = \frac{1}{\hbar\omega} \left[\int_0^\infty dt'' \langle \psi_0 | [J_j(t + t''), J_i(t)] | \psi_0 \rangle E_j e^{+i\omega t''} \right] E_j e^{-i\omega t} \quad (1.131)$$

Inside the commutator, we will use that fact that the system is translational invariant, and write,

$$\langle J_i(t) \rangle = \frac{1}{\hbar\omega} \left[\int_0^\infty dt'' \langle \psi_0 | [J_j(t''), J_i(0)] | \psi_0 \rangle E_j e^{+i\omega t''} \right] E_j e^{-i\omega t} \quad (1.132)$$

This tells us that the current oscillates with the same frequency as the applied field, to the lowest order. We can define the conductivity,

$$\sigma_{ij}(\omega) = \frac{1}{\hbar\omega} \int_0^\infty dt \langle \psi_0 | [J_j(t), J_i(0)] | \psi_0 \rangle \quad (1.133)$$

This is the Kubo formula for conductivity.

Let us do a little bit more manipulations. Use,

$$\mathbf{J}(t) = e^{\frac{i}{\hbar} H_0 t} \mathbf{J}(0) e^{-\frac{i}{\hbar} H_0 t} \quad (1.134)$$

and insert 1,

$$\sigma_{xy}(\omega) = \frac{1}{\hbar\omega} \int_0^\infty dt e^{i\omega t} \sum_n \left[e^{\frac{i}{\hbar}(E_0 - E_n)t} \langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle - e^{-\frac{i}{\hbar}(E_0 - E_n)t} \langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle \right] \quad (1.135)$$

Notice that the $n = 0$ terms cancel. We can do the t integral by shifting $\omega \rightarrow \omega + i\epsilon$, and get,

$$\sigma_{xy}(\omega) = -\frac{i}{\omega} \sum_{n \neq 0} \left[\frac{\langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle}{\hbar\omega + E_n - E_0} - \frac{\langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle}{\hbar\omega + E_0 - E_n} \right] \quad (1.136)$$

If we want to send $\omega \rightarrow 0$ now, it seems that there is a divergence. If we expand the denominators as,

$$\frac{1}{\hbar\omega + E_n - E_0} = \frac{1}{E_n - E_0} - \frac{\hbar\omega}{(E_n - E_0)^2} + \mathcal{O}(\omega^2) \quad (1.137)$$

Then, the first term along with the $1/\omega$ in front will make a divergent term. But for hall conductivity, this term can be shown to vanish (I do not know how!). Therefore the conductivity in the DC limit is,

$$\sigma_{xy}(\omega) = i\hbar \sum_{n \neq 0} \frac{\langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle - \langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle}{(E_n - E_0)^2} \quad (1.138)$$

We will use this formula now.

Back to Hall Conductivity on the Torus

Let me recap the story. We took our sample to be a torus, and on top of the magnetic field, we add two fluxes, each along the cycles of the torus. This is our cute little trick to apply an emf. Then, we had the perturbation term,

$$\Delta H = - \sum_{i=x,y} \frac{J_i \Phi_i}{L_i} \quad (1.139)$$

Due to this perturbation, the ground state will get shifted. Let us assume non degenerate ground state. Then we know from our quantum mechanics classes,

$$\begin{aligned} |\psi'_0\rangle &= |\psi_0\rangle + \sum_{n \neq 0} |\psi_n\rangle \frac{\langle \psi_n | \Delta H | \psi_0 \rangle}{E_n - E_0} \\ &= |\psi_0\rangle - \sum_{n \neq 0} |\psi_n\rangle \sum_{i=x,y} \frac{\Phi_i}{L_i} \frac{\langle \psi_n | J_i | \psi_0 \rangle}{E_n - E_0} \end{aligned} \quad (1.140)$$

For an infinitesimal change of flux,

$$\frac{|\psi'_0\rangle - |\psi_0\rangle}{\Phi_i} \equiv \frac{\partial |\psi_0\rangle}{\partial \Phi_i} = -\frac{1}{L_i} \sum_{n \neq 0} |\psi_n\rangle \frac{\langle \psi_n | J_i | \psi_0 \rangle}{E_n - E_0} \quad (1.141)$$

We can then find the Hall conductivity using the Kubo formula. We change the fluxes to produce EMF, and look for the transverse conductivity, and the Kubo formula gives us

precisely this for any general current. We have⁸,

$$\begin{aligned}
\sigma_{xy}(\omega) &= i\hbar L_x L_y \sum_{n \neq 0} \frac{\langle \psi_0 | J_y | \psi_n \rangle \langle \psi_n | J_x | \psi_0 \rangle - \langle \psi_0 | J_x | \psi_n \rangle \langle \psi_n | J_y | \psi_0 \rangle}{(E_n - E_0)^2} \\
&= i\hbar \left[\left\langle \frac{\partial \psi_0}{\partial \Phi_y} \middle| \frac{\partial \psi_0}{\partial \Phi_x} \right\rangle - \left\langle \frac{\partial \psi_0}{\partial \Phi_x} \middle| \frac{\partial \psi_0}{\partial \Phi_y} \right\rangle \right] \\
&= i\hbar \left[\frac{\partial}{\partial \Phi_y} \langle \psi_0 | \frac{\partial}{\partial \Phi_x} | \psi_0 \rangle - \frac{\partial}{\partial \Phi_x} \langle \psi_0 | \frac{\partial}{\partial \Phi_y} | \psi_0 \rangle \right] \tag{1.142}
\end{aligned}$$

This isn't particularly illuminating in the first glance. What has it to do with topology at all? What we have above is just the variation of the ground state as the flux is changing. It turns out, this is a geometric quantity! Let us take a step back and understand the geometry lurking behind the above expression.

1.3.1 The Berry Phase

Let us consider a general Hamiltonian,

$$H(x^a; \lambda_i) \tag{1.143}$$

Here, x^a are the dynamical variables, and λ^i are the parameters of the Hamiltonian. These are usually things that we can control. Like in the example we were stuck, the flux was something that in our control.

Now, we do the following : We pick a particular set of values for the parameters λ , and prepare the system in an energy eigen state $|\psi\rangle$. Then, we “slowly” change the parameters. How slow is slow would depend on the problem at hand. But we will ensure that we are slow enough to ensure that the adiabatic theorem is valid. This theorem says that if we change the parameters of the Hamiltonian sufficiently slowly, then if we started with say the n th eigen state of $H(\lambda_0)$, (assume it is non degenerate), the system will stay at the n th eigen state of $H(\lambda)$.

There are several caveats. The first one is, what if this change of parameters result in a level crossing? Level crossing means the energy eigen states cross each other. We will assume that this is not happening. The second caveat is if the energy eigen state we started with has degeneracy. This case, we will consider later and will result in non-abelian Berry phase.

⁸I do not understand why there is a factor of area suddenly appearing.

What happens if we change the parameters in such a way that after exploring some region of the parameter space, we get back to the original starting values? The adiabatic theorem says that it has to be in the same energy eigen state. But, we can have a shift in the phase,

$$|\psi\rangle \rightarrow e^{i\gamma} |\psi\rangle \quad (1.144)$$

This phase, is actually not a trivial one. We can actually measure this. All we have to do is to have two states, and send one of them in the journey while the other waits. This phase, will have a contribution from the Hamiltonian, $e^{-\frac{i}{\hbar}Et}$. We will ignore this. We will focus on the phase that is coming purely from the change of parameters.

Let us find this phase. We know, the state has to obey the Schrodinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H(\lambda(t)) |\psi\rangle = 0 \quad (1.145)$$

We choose the energy eigen state to have zero energy (let it be the ground state). This way, we can focus purely on the phase due to the parameter change. Now, as we change the parameters, the state $|\psi\rangle$ will slowly change to the ground state of $H(\lambda)$. For each λ , let us choose a reference ground state,

$$|0(\lambda)\rangle : H |0(\lambda)\rangle = 0 \quad (1.146)$$

There is nothing special about this choice. We could have chosen some other reference state,

$$|0'(\lambda)\rangle = e^{i\omega(\lambda)} |0(\lambda)\rangle \quad (1.147)$$

We will see what this change of choice will do soon. But now, we know from the adiabatic theorem that,

$$|\psi(t)\rangle = U(t) |0(\lambda(t))\rangle \quad (1.148)$$

Here, $U(t)$ is just a phase. Plug this into the Schrodinger equation, and multiply by $\langle\psi(t)|$ from the left,

$$U^*(t) \dot{U}(t) + \langle 0(\lambda(t)) | \frac{\partial}{\partial t} | 0(\lambda(t)) \rangle = 0 \quad (1.149)$$

We have,

$$\langle 0(\lambda(t)) | \frac{\partial}{\partial t} | 0(\lambda(t)) \rangle = \langle 0(\lambda(t)) | \frac{\partial}{\partial \lambda^i} | 0(\lambda(t)) \rangle \dot{\lambda}^i \quad (1.150)$$

Define the Berry connection,

$$\mathcal{A}_i = -i \langle 0(\lambda) | \frac{\partial}{\partial \lambda^i} | 0(\lambda) \rangle \quad (1.151)$$

Then we have,

$$\dot{U}(t) = -i\mathcal{A}_i \dot{\lambda}^i U(t) \quad (1.152)$$

This can be solved easily to obtain,

$$U(t) = \exp\left\{-i \int dt \mathcal{A}_i \dot{\lambda}^i\right\} \quad (1.153)$$

Now, if we are taking a closed path C in the parameter space, then we have,

$$e^{i\gamma} = \exp\left\{-i \oint_C d\lambda^i \mathcal{A}_i\right\} \quad (1.154)$$

This is the Berry phase.

Now, let us understand what happens if we chose a different reference state for each λ . For example, if we chose,

$$|0'(\lambda)\rangle = e^{i\omega(\lambda)} |0(\lambda)\rangle \quad (1.155)$$

then,

$$\mathcal{A}'_i = -i \langle 0'(\lambda) | \frac{\partial}{\partial \lambda^i} | 0'(\lambda) \rangle = \mathcal{A}_i + \frac{\partial \omega}{\partial \lambda^i} \quad (1.156)$$

This is exactly a gauge transformation! Different choices of phases for the reference states result in a corresponding gauge transformation in the Berry connection. And, the Berry phase, which is an observable, do not depend on the gauge. We can see this easily because,

$$\oint_C d\lambda^i \frac{\partial \omega}{\partial \lambda^i} = 0 \quad (1.157)$$

We can, in analogy with gauge theory define a gauge invariant field strength,

$$\mathcal{F}_{ij} = \frac{\partial \mathcal{A}_i}{\partial \lambda^j} - \frac{\partial \mathcal{A}_j}{\partial \lambda^i} \quad (1.158)$$

Then, by Stokes's theorem,

$$e^{i\gamma} = \exp\left\{-i \oint_C d\lambda^i \mathcal{A}_i\right\} = \exp\left\{-i \int_S dS^{ij} \mathcal{F}_{ij}\right\} \quad (1.159)$$

where S is the surface in the parameter space which is bounded by the curve C .

Once Again, Back to Hall Conductivity on a Torus

We have the fluxes on the torus to help generate EMF. As we know already, the wave functions are sensitive to the flux only if Φ/Φ_0 is not an integer. When Φ changes from 0 to Φ_0 , the energy levels are the same, except we have a spectral flow. This means the fluxes, which are our parameters, are periodic. If we change the fluxes slowly from 0 to Φ_0 , then this is equivalent to taking a loop in parameter space. That means, the flux parameter space is also a torus, let us call it T_Φ^2 . To have nice periodicity on this flux torus, let us define,

$$\theta_i = \frac{2\pi\Phi_i}{\Phi_0} \quad (1.160)$$

Therefore, changing flux from zero to Φ_0 is equivalent to changing θ from 0 to 2π . Then, we can define the Berry connection,

$$\mathcal{A}_i(\Phi) = -i \langle \psi_0 | \frac{\partial}{\partial \theta_i} | \psi_0 \rangle \quad (1.161)$$

Then, our conductivity is,

$$\begin{aligned} \sigma_{xy}(\omega) &= i\hbar \left[\frac{\partial}{\partial \Phi_y} \langle \psi_0 | \frac{\partial}{\partial \Phi_x} | \psi_0 \rangle - \frac{\partial}{\partial \Phi_x} \langle \psi_0 | \frac{\partial}{\partial \Phi_y} | \psi_0 \rangle \right] \\ &= -\frac{e^2}{\hbar} \left(\frac{\partial \mathcal{A}_x}{\partial \theta^y} - \frac{\partial \mathcal{A}_y}{\partial \theta^x} \right) \\ &= -\frac{e^2}{\hbar} \mathcal{F}_{xy} \end{aligned} \quad (1.162)$$

We see that the Hall conductivity is purely a geometric quantity! How do we see that this is quantized? For that, we have to consider the average over the fluxes. If we take an average over the fluxes,

$$\bar{\sigma}_{xy} = -\frac{e^2}{\hbar} \frac{1}{\Phi_0^2} \int d\Phi_x d\Phi_y \mathcal{F}_{xy} = -\frac{e^2}{2\pi\hbar} \frac{1}{2\pi} \int d^2\theta \mathcal{F}_{xy} \quad (1.163)$$

We define something called the first Chern number,

$$C = \frac{1}{2\pi} \int d^2\theta \mathcal{F}_{xy} \quad (1.164)$$

The Chern number, it turns out is an integer. Therefore, we have,

$$\bar{\sigma}_{xy} = -\frac{e^2}{2\pi\hbar} C \quad \text{where } C \in \mathbb{Z} \quad (1.165)$$

To understand why the Chern number is an integer, we do quantum mechanics on the flux

torus, T_Φ^2 . This is slightly confusing, so brace yourselves. Consider a wave function on the flux torus,

$$\psi(\theta_x, \theta_y) \quad (1.166)$$

We have a “magnetic field”,

$$B = \mathcal{F}_{xy} \quad (1.167)$$

Since we have gauge symmetry, let me choose $\mathcal{A}_x = 0$ and $\mathcal{A}_y = -\theta_x B$. For this choice, we must assume that the magnetic field is constant. Let us do so for simplicity. But remember, on the torus, $\mathcal{F}$ is closed, but cannot be exact. Then it seems absurd to write down the Berry connection. But the statement is, $\mathcal{F}$ cannot be uniquely written as the exterior derivative of a connection. Since our connection can have gauge redundancy, the non uniqueness is not our concern. As we see, the gauge potential I wrote down is not single valued. We just have to make sure that quantum mechanics is alright, i.e, the wave functions have well good boundary conditions. For that, we do the following : on the flux torus, we go around the θ_x direction, then we have,

$$\psi(\theta_x + 2\pi, \theta_y) = \psi(\theta_x, \theta_y) \quad (1.168)$$

This is simply because we chose the gauge. Now, going around θ_y direction gives,

$$\psi(\theta_x, \theta_y + 2\pi) = e^{-i\theta_x B 2\pi} \psi(\theta_x, \theta_y) \quad (1.169)$$

Now, we go around θ_x and θ_y , and in the reverse order. So, we have,

$$\psi(\theta_x, \theta_y) \xrightarrow{\theta_x \rightarrow \theta_x + 2\pi} \psi(\theta_x + 2\pi, \theta_y) \xrightarrow{\theta_y \rightarrow \theta_y + 2\pi} e^{i\theta_x B 2\pi} \psi(\theta_x + 2\pi, \theta_y + 2\pi) \quad (1.170)$$

and,

$$\psi(\theta_x, \theta_y) \xrightarrow{\theta_y \rightarrow \theta_y + 2\pi} e^{i\theta_x B 2\pi} \psi(\theta_x, \theta_y + 2\pi) \xrightarrow{\theta_x \rightarrow \theta_x + 2\pi} e^{i(\theta_x + 2\pi) B 2\pi} \psi(\theta_x + 2\pi, \theta_y + 2\pi) \quad (1.171)$$

For the wave function to have good boundary conditions, we must have,

$$4\pi^2 B = 2\pi C \quad \text{where } C \in \mathbb{Z} \quad (1.172)$$

Therefore,

$$\frac{1}{2\pi} \int d^2\theta B = \frac{1}{2\pi} (2\pi)^2 \times \frac{2\pi C}{(2\pi)^2} = C \in \mathbb{Z} \quad (1.173)$$

I have not understood yet how to write a general argument for non constant $\mathcal{F}$. I will write it when I understand this. One way to prove is to somehow prove that chern number do not change continuously.

1.4 Hall Effect on a Lattice

So far, we described Hall conductivity on a sheet (actually, we cheated and used an annular ring for the main argument) and on a torus. In both cases, topology played a crucial role. Now, we turn to a periodic lattice. On a periodic lattice, say period a along x and b along y , the momenta are restricted to take discrete values, namely,

$$-\frac{\pi}{a} \leq k_x \leq \frac{\pi}{a} \quad \text{and} \quad -\frac{\pi}{b} \leq k_y \leq \frac{\pi}{b} \quad (1.174)$$

This is called the Brillouin zone. Importantly, the value of k at the ends of the Brillouin zone are identified. Therefore, we have a torus in the $k_x - k_y$ space! This is a topologically interesting space, and we are going to exploit its robustness once again.

The discussion that is going to follow will not assume a specific Hamiltonian. We know already that if we know that there is a periodic potential, the wave functions are going to be Bloch waves. i.e, The wave functions will be of the form,

$$\psi_{\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{x}} u_{\mathbf{k}}(\mathbf{x}) \quad (1.175)$$

and $u_{\mathbf{k}}(\mathbf{x})$ is periodic on the unit cell, that is, $u_{\mathbf{k}}(\mathbf{x} + \mathbf{e}) = u_{\mathbf{k}}(\mathbf{x})$ if $\mathbf{e} = (a, 0)$ or $(0, b)$.

We will assume the following criteria on the lattice.

- As usual, the electrons are non interacting. Only Pauli exclusion principle stands in the way of particles.
- The single particle spectrum forms bands, and each band has states labelled by $\mathbf{k}$.
- We will assume that we have an insulator. i.e, there is a gap between bands, and the Fermi energy is in this gap. Therefore, the band above Fermi energy is empty.

1.4.1 TKNN Formula for Hall Conductance on a Lattice

With a Hamiltonian with the above properties on a periodic lattice, we know that the momentum space forms a torus. Therefore, we can define a Berry connection on this torus (by considering $\mathbf{k}$ as the parameter we vary),

$$\mathcal{A}_i(\mathbf{k}) = -i \langle u_{\mathbf{k}} | \frac{\partial}{\partial k^i} | u_{\mathbf{k}} \rangle \quad (1.176)$$

Then, we have the Berry curvature,

$$\mathcal{F}_{xy} = \frac{\partial \mathcal{A}_x}{\partial k^y} - \frac{\partial \mathcal{A}_y}{\partial k^x} = -i \left[\left\langle \frac{\partial u_{\mathbf{k}}}{\partial k^y} \middle| \frac{\partial u_{\mathbf{k}}}{\partial k^x} \right\rangle - \left\langle \frac{\partial u_{\mathbf{k}}}{\partial k^x} \middle| \frac{\partial u_{\mathbf{k}}}{\partial k^y} \right\rangle \right] \quad (1.177)$$

Then, we can find the first Chern number,

$$C = -\frac{1}{2\pi} \int_{T^2} d^2k \mathcal{F}_{xy} \in \mathbb{Z} \quad (1.178)$$

The Chern number is, once again an integer⁹. Also, this is a topological invariant (I do not know how to prove this!), and in this context called the TKNN invariant. Therefore for each band, we can find the Chern number, call it C_α for band α .

From our experience from the Hall conductivity on a torus, we know that the Berry curvature is related to the Hall conductivity. In the case of the torus, we got the quantized values of the Hall conductivity once we averaged over the flux torus. Averaging the Berry connection over the flux torus produced the integer C which is the Chern number. So, we might be inclined to conjecture that for a lattice, the Hall conductivity is,

$$\sigma_{xy} = \frac{e^2}{2\pi\hbar} \sum_{\alpha} C_{\alpha} \quad (1.179)$$

where the sum is over the filled bands. This turns out to be correct and is called the TKNN formula.

Proof of TKNN Formula

We will start with Kubo formula. The state we start with ($|\psi_0\rangle$ in our Kubo formula), we take it to be the multi particle state corresponding to a filled band. Since we are assuming that electrons are not interacting with each other, this multi particle state is just tensor products of single particle states, corresponding to the band α and momentum $\mathbf{k}$. The final state, is any, except the states from the filled band. In other words, states from the unfilled band. One thing to note is that the initial state is actually,

$$|\psi_0\rangle = \sum_{\alpha: E_{\alpha} < \mathcal{E}_F} \int_{T^2} \frac{d^2k}{(2\pi)^2} |u_{\mathbf{k}}^{\alpha}\rangle \quad (1.180)$$

⁹To discover this once again, we do what we know : QM on the momentum torus. We will consider wave functions of the form $\psi(k_x, k_y)$ and proceed as before.

Then, we can write the Kubo formula,

$$\sigma_{xy} = i\hbar \sum_{\alpha: E_\alpha < \mathcal{E}_F} \int_{T^2} \frac{d^2k}{(2\pi)^2} \sum_{\beta: E_\beta > \mathcal{E}_F} \int_{T^2} \frac{d^2k'}{(2\pi)^2} \frac{\langle u_{\mathbf{k}}^\alpha | J_y | u_{\mathbf{k}'}^\beta \rangle \langle u_{\mathbf{k}'}^\beta | J_x | u_{\mathbf{k}}^\alpha \rangle - \langle u_{\mathbf{k}}^\alpha | J_x | u_{\mathbf{k}'}^\beta \rangle \langle u_{\mathbf{k}'}^\beta | J_y | u_{\mathbf{k}}^\alpha \rangle}{(E_\beta(\mathbf{k}') - E_\alpha(\mathbf{k}))^2} \quad (1.181)$$

To evaluate this, we need to know what the current is. Roughly speaking, since we have bands, the energies depend on the momentum. Therefore, we can use the group velocity to define a current. To see this, let us start with the Schrodinger equation,

$$H |\psi_{\mathbf{k}}\rangle = E(\mathbf{k}) |\psi_{\mathbf{k}}\rangle \implies e^{-i\mathbf{k}\cdot\mathbf{x}} H e^{i\mathbf{k}\cdot\mathbf{x}} |u_{\mathbf{k}}\rangle = E(\mathbf{k}) |u_{\mathbf{k}}\rangle \quad (1.182)$$

That is, the momentum dependent spectrum of $|u_{\mathbf{k}}\rangle$ is computed by the operator,

$$\tilde{H}(\mathbf{k}) = e^{-i\mathbf{k}\cdot\mathbf{x}} H e^{i\mathbf{k}\cdot\mathbf{x}} \quad (1.183)$$

It is better to define the current using $\tilde{H}$ because this operator have $u_{\mathbf{k}}$'s as its eigen states. Define the current to be $e \times$ group velocity. i.e,

$$\mathbf{J} = \frac{e}{\hbar} \frac{\partial \tilde{H}}{\partial \mathbf{k}} \quad (1.184)$$

We can show that this definition is the same as the usual definition, $\mathbf{J} = e\dot{\mathbf{x}}$ if the Hamiltonian is the usual $H = \frac{(\mathbf{p} + e\mathbf{A})^2}{2m}$ ¹⁰. Sandwiching between $e^{-i\mathbf{k}\cdot\mathbf{x}}$ is like translating in momentum space by $\hbar\mathbf{k}$. This gives us,

$$\tilde{H}(\mathbf{k}) = \frac{(\mathbf{p} + \hbar\mathbf{k} + e\mathbf{A})^2}{2m} \quad (1.185)$$

Differentiating this with $\mathbf{k}$, we have mechanical momentum by mass, which is just $\dot{\mathbf{x}}$. Therefore, the current is just $\mathbf{J} = e\dot{\mathbf{x}}$.

Now, let us roll up our sleeves and do the algebra. We have to find,

$$\begin{aligned} \langle u_{\mathbf{k}'}^\beta | \frac{\partial \tilde{H}}{\partial k^i} | u_{\mathbf{k}}^\alpha \rangle &= \langle u_{\mathbf{k}'}^\beta | \frac{\partial \tilde{H}}{\partial k^i} | u_{\mathbf{k}}^\alpha \rangle \\ &= \langle u_{\mathbf{k}'}^\beta | \frac{\partial}{\partial k^i} (\tilde{H} | u_{\mathbf{k}}^\alpha \rangle) - \langle u_{\mathbf{k}'}^\beta | \tilde{H} | \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k^i} \rangle \\ &= \langle u_{\mathbf{k}'}^\beta | \frac{\partial}{\partial k^i} (E_\alpha(\mathbf{k}) | u_{\mathbf{k}}^\alpha \rangle) - \langle u_{\mathbf{k}'}^\beta | \tilde{H} | \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k^i} \rangle \\ &= (E_\alpha(\mathbf{k}) - E_\beta(\mathbf{k}')) \langle u_{\mathbf{k}'}^\beta | \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k^i} \rangle \end{aligned} \quad (1.186)$$

¹⁰Remember, for the lattice, we are not even prescribing a Hamiltonian. This is just a sanity check!

The term with $\langle u^\alpha | u^\beta \rangle$ is zero because α and β are different bands, one filled and the other not filled. Similarly,

$$\langle u_{\mathbf{k}}^\alpha | \frac{\partial \tilde{H}}{\partial k^i} | u_{\mathbf{k}'}^\beta \rangle = -(E_\alpha(\mathbf{k}) - E_\beta(\mathbf{k}')) \left\langle \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k^i} | u_{\mathbf{k}'}^\beta \right\rangle \quad (1.187)$$

Plugging this into Kubo formula gives us,

$$\sigma_{xy} = \frac{ie^2}{\hbar} \sum_{\alpha, \beta} \int_{T^2} \frac{d^2 k}{(2\pi)^2} \frac{d^2 k'}{(2\pi)^2} \left[\left\langle \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_y} | u_{\mathbf{k}'}^\beta \right\rangle \left\langle u_{\mathbf{k}'}^\beta | \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_x} \right\rangle - \left\langle \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_x} | u_{\mathbf{k}'}^\beta \right\rangle \left\langle u_{\mathbf{k}'}^\beta | \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_y} \right\rangle \right] \quad (1.188)$$

Now, we can use the completeness relation to write,

$$\sum_{\beta} \int_{T^2} \frac{d^2 k'}{(2\pi)^2} | u_{\mathbf{k}'}^\beta \rangle \langle u_{\mathbf{k}'}^\beta | = \mathbf{1} - \sum_{\alpha} \int_{T^2} \frac{d^2 k}{(2\pi)^2} | u_{\mathbf{k}}^\alpha \rangle \langle u_{\mathbf{k}}^\alpha | \quad (1.189)$$

If we plug this into the conductivity, only the terms coming from the identity survives! It is easy to see this. For example, we would take a term of the form $\left\langle \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_x} | u_{\mathbf{k}'}^\beta \right\rangle$ and integrate by parts. Since there are two integration by parts happening for say the first term, there is no sign change. After this integration by parts, the first term becomes the same as the second term, and hence they cancel. Therefore, we are now left with only a sum over the filled bands,

$$\sigma_{xy} = \frac{ie^2}{\hbar} \sum_{\alpha} \int_{T^2} \frac{d^2 k}{(2\pi)^2} \left[\left\langle \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_y} | \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_x} \right\rangle - \left\langle \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_x} | \frac{\partial u_{\mathbf{k}}^\alpha}{\partial k_y} \right\rangle \right] \quad (1.190)$$

What is sitting inside the integral is the Berry curvature, times a factor of i . Integrating this over the space of momenta gives us the Chern number. Hence, we arrive at the proof of the TKNN formula,

$$\sigma_{xy} = \frac{e^2}{2\pi\hbar} \sum_{\alpha} C_{\alpha} \quad (1.191)$$

This is remarkable! We see that the conductivity of a lattice is a topological property. If we change the lattice in such a way that we maintain the assumptions, the Hall conductivity cannot change. This is because the Chern numbers are topological and will not change unless there is a change in the topology (in this case the topology of the bands).

1.4.2 Example : Chern Insulator

The name of the example is counter intuitive! How can an insulator have any conductance at all? We will see now that even a model with filled bands, and even no magnetic field gives

rise to a Hall response.

We consider a model with two bands. The Hamiltonian is given by,

$$\tilde{H}(\mathbf{k}) = \vec{E}(\mathbf{k}) \cdot \vec{\sigma} + \epsilon(\mathbf{k}) \quad (1.192)$$

Here, $\vec{E}$ is a vector, and not the electric field. As before, $\mathbf{k} \in T^2$. The spectrum of this Hamiltonian is easy to get, and are given by the energies $\epsilon(\mathbf{k}) \pm |\vec{E}(\mathbf{k})|$. We see that the system has two bands (assuming $|\vec{E}(\mathbf{k})| \neq 0$ for all values of $\mathbf{k}$).

We can now employ the TKNN formula. Naively, we expect the Chern numbers for the bands to be zero, and therefore there is no conductivity. But, we will now see that we can cook up situations in which the Chern numbers are non zero!

To do this, let us interpret the Chern numbers in a more geometric way. We know that the Chern numbers are topological. They measure the non-exactness of a closed two form. To elaborate, if we have a closed two form $d\mathcal{F} = 0$, then if $\mathcal{F} = d\mathcal{A}$ for a unique $\mathcal{A}$, we will call the form $\mathcal{F}$ exact. If it is exact, the integral of it over a closed two dimensional manifold will vanish due to the Stokes theorem. Here, our two form of interest is the Berry curvature. If the Berry curvature is non exact, then the Chern number - which is the integral of the Berry curvature around the two dimensional manifold - will be non zero. For the case of the Chern insulator, the closed two dimensional manifold is the torus in the momentum space. From geometry, we know that the de Rham cohomology group of the torus is $H_2(T^2) \cong \mathbb{Z}$ ¹¹. That means, there is exactly one two form which is closed and non exact for each integer. This integer marks the number of times the form “wraps around” the torus. What this means is that we do not have to sweat to find the Berry curvature! Find one two form that wraps around the torus, and the Berry curvature is equivalent to this two form - which means it differs from the form only by an exact form - and hence the Chern number can be found by integrating this form!

To find a two form that wraps around the torus, we pretty much have only one place to look at : $\vec{E}(\mathbf{k})$ in the Hamiltonian. This is the only set of parameters that we have in our hand to tune. Define a normalized vector,

$$\vec{n}(\mathbf{k}) = \frac{\vec{E}(\mathbf{k})}{|\vec{E}(\mathbf{k})|} \quad (1.193)$$

¹¹Alternatively think of the homology group, and use de Rham’s theorem

This unit vector is a map that takes $T^2 \rightarrow S^2$. If we sweep through T^2 , $\vec{n}$ may sweep through the sphere multiple times. [It is not clear to me how David Tong finds the two form. I am just quoting the result here!](#) We can write the Chern number as,

$$C = \frac{1}{4\pi} \int_{T_{\mathbf{k}}^2} d^2k \, \vec{n}(\mathbf{k}) \cdot \left(\frac{\partial \vec{n}}{\partial k_x} \times \frac{\partial \vec{n}}{\partial k_y} \right) \quad (1.194)$$

The integral gives the area of the sphere swept when we sweep the torus, 4π normalises it. Therefore, C is the number of times the sphere is wrapped around by the map.

An example is the Hamiltonian,

$$\tilde{H}(\mathbf{k}) = \sin k_x \sigma_1 + \sin k_y \sigma_2 + (m + \cos k_x + \cos k_y) \sigma_3 \quad (1.195)$$

Here $|\vec{E}(\mathbf{k})|$ is not zero in general for all $\mathbf{k}$. But, if $m = 0$, this gap is zero for $(k_x, k_y) = (\pm\pi, \mp\pi)$, and is no longer an insulator, also the gap closes for $m = \pm 2$ as well. The Chern number in the regions where the gap is open are different from zero and can be found from direct calculation. I am skipping that here.

1.5 Chern-Simons Theory

Interestingly, the low energy behaviour of the quantum Hall ground state can be described using a field theory! This is what we will describe now. First, let us set the stage. We will no longer talk about individual electrons or their wavefunctions.

First, to ensure that we are working with the quantum Hall ground state, we will assume that the energies we are interested in are lower than the band gap¹². Therefore, if we perturb the system, no electron will jump to the next band and create a new degree of freedom that we haven't accounted for. We will assume that there are no other degrees of freedom other than the electromagnetic field in the lowest band.

Next, we will not treat the electromagnetic field $A_\mu(x)$ as a quantum field. In all our descriptions of Hall effect, the electromagnetic field was a parameter that we tuned to extract the Hall conductivity. The same stays true here. Also, the initial electric and magnetic fields - i.e, a $\mathbf{B}$ field along $\hat{z}$ and no $\mathbf{E}$ field - will not be included in A_μ . Instead, A_μ will denote the perturbations on top of this existing field. We will also assume that these perturbations

¹²The band gap should not be confused with the Landau levels. At each band, there can be many Landau levels.

are “small” such that the Hall state stays the same.

Our material is in a $2 + 1$ dimensional slice of space time. J_μ the current is what couples to the electromagnetic field. The coupling is,

$$S_A = \int d^3x A_\mu(x) J^\mu(x)$$

This term is gauge invariant, because $\partial_\mu J^\mu(x) = 0$. Now, we want to compute the partition function. This is,

$$Z[A_\mu(x)] = \int \mathcal{D}(\text{Fields}) \exp \left\{ \frac{i}{\hbar} S[\text{Fields}; A_\mu] \right\} = e^{\frac{i}{\hbar} S_{\text{eff}}[A_\mu(x)]} \quad (1.196)$$

Here $S_{\text{eff}}[A]$ is called the effective action. Since we do not know the microscopic action, there is no way to actually evaluate the effective action. Our strategy instead will be to just write it down, based on symmetry arguments. Once we write it down, the current expectation value is,

$$\langle J^\mu(x) \rangle = \frac{\delta S_{\text{eff}}[A]}{\delta A_\mu(x)} \quad (1.197)$$

The effective action will be assumed to be local. This is because we are describing low energy physics. The length scale $\frac{v\hbar}{\Delta E}$ where v is the characteristic velocity and ΔE is the band gap will be assumed to be small. The next assumption will be that the action is gauge invariant.

To construct a gauge invariant action, the best approach will be to use $\mathbf{E}$ and $\mathbf{B}$. Since we are interested in small $\mathbf{E}$ and $\mathbf{B}$ ¹³, we will be interested in terms with lowest number of powers of $\mathbf{E}$ and $\mathbf{B}$. Also, we will look at fields that vary only on long distance scales, we will be interested in fewer derivatives on the fields as well. The simplest terms will be of quadratic order in $\mathbf{E}$ and $\mathbf{B}$. We will not consider linear terms because of rotational invariance.

In three dimensions, there is another term, known as the Chern-Simons term. This is,

$$S_{\text{CS}}[A] = \frac{k}{4\pi} \int d^3x \epsilon^{\mu\nu\sigma} A_\mu(x) \partial_\nu A_\sigma(x) \quad (1.198)$$

The existence of this term owes to $\epsilon^{\mu\nu\sigma}$. k is called the level of the Chern-Simons term.

¹³remember, $\mathbf{E}$ and $\mathbf{B}$ are the perturbations not the full fields.

Now, we should verify if the Chern-Simons term is gauge invariant. It is not obvious from the action itself. If we make the change $A_\mu + \partial_\mu \omega$, then the action changes as,

$$\delta S_{\text{CS}}[A] = \frac{k}{4\pi} \int d^3x \partial_\mu (\epsilon^{\mu\nu\sigma} \omega \partial_\nu A_\sigma)$$

The change is a total derivative. If for some reason we cannot throw away the total derivative term, then it seems like the action is not gauge invariant, and we would have to have $k = 0$. But remember, the important quantity that should be gauge invariant is $e^{\frac{i}{\hbar} S_{\text{CS}}[A]}$. Therefore, we are okay if the above term is actually an integer times 2π . We will now see that this is indeed the case, and hence will reproduce Hall conductivity.

The next thing that we observe is that the Chern-Simons term does not obey parity. If we change $x^1 \rightarrow -x^1$ while keeping the other coordinates the same, then A_1 and ∂_1 flip sign, which therefore flips the sign of the action. But this is okay, since the Hall system does not obey parity, due to the magnetic field that is on all times.

1.5.1 Quantization of the Chern-Simons Level

We will work with finite temperatures. The easiest way to achieve this is to use Euclidean signature and make the Euclidean time periodic with period $\beta = \frac{1}{T}$. We can, at the end, take a zero temperature limit if needed.

Periodic time brings a novelty. We can consider non-singular gauge transformations. We know that the gauge transformations only have to leave the fields (which we integrated over) the same. Therefore, a gauge transformation ω such that $e^{i e \omega / \hbar}$ is single valued is fine. For example,

$$\omega = \frac{2\pi \hbar \tau}{e\beta}$$

is a valid gauge transformation. These are known as large gauge transformations, which cannot be continuously deformed to the identity. The zero mode of A_0 under this transformation changes as,

$$A_0 \rightarrow A_0 + \frac{2\pi \hbar}{e\beta}$$

The above two are equivalent field configurations. In other words, the zero mode of A_0 is periodic with period $\frac{2\pi \hbar}{e\beta}$.

We will also change the geometry of the spatial slice. We will assume it to be a sphere, or any compact two dimensional manifold. Let us take A_0 to be a constant. Then, the Chern-Simons action is,

$$S_{\text{CS}}[A] = \frac{k}{4\pi} \int_{S^2 \times S^1} [A_0 F_{12} + A_1 F_{20} + A_2 F_{01}]$$

At first sight, it seems like we can drop the last two terms. But this is incorrect. If we assume that there is a non zero flux through the sphere S^2 , then that means there is a magnetic monopole sitting at the center of the sphere. Then, A_i would have a singular expression at the center of the sphere, hence discarding the F_{0i} terms that multiply the A_i 's will be too quick.

To evaluate this, let us denote $\frac{2\pi\hbar}{e} = s$, so that $\delta A_0 = \frac{s}{\beta}$. Then,

$$\delta S_{\text{CS}}[A] = \frac{k}{4\pi} \int_{S^2 \times S^1} \epsilon^{\mu\nu\sigma} \delta A_\mu F_{\nu\sigma}$$

The change in the action is gauge invariant since δA and F are both gauge invariant. For our large gauge transformations,

$$\delta A_\mu = \frac{s}{\beta} \delta_{\mu 0}$$

Hence,

$$\frac{\partial \delta S_{\text{CS}}}{\partial s} = \frac{k}{4\pi\beta} \int_{S^2 \times S^1} (F_{12} - F_{21}) = \frac{k}{2\pi} \int_{S^2} F_{12}$$

We will assume that the magnetic flux is due to one single quantum of flux, therefore,

$$\frac{1}{2\pi} \int_{S^2} F_{12} = \frac{\hbar}{e}$$

Therefore,

$$\frac{\partial \delta S_{\text{CS}}}{\partial s} = \frac{\hbar k}{e}$$

If we use the fact that $s = 0 \implies S_{\text{CS}} = 0$ - i.e, no gauge transformation keeps the action at zero - we can integrate this differential equation and get,

$$\delta S_{\text{CS}} = \frac{\hbar k s}{e} = \frac{2\pi\hbar^2 k}{e^2} \tag{1.199}$$

For the theory to be gauge invariant, we have to demand,

$$\exp\left\{\frac{i}{\hbar} \delta S_{\text{CS}}\right\} = 1$$

Therefore, we must have,

$$\frac{\hbar k}{e^2} \in \mathbb{Z} \iff k = \frac{e^2 \nu}{\hbar} \quad (1.200)$$

We can now look at the conductivity. We have,

$$J_i = \frac{\delta S_{\text{CS}}[A]}{\delta A_i} = \frac{k}{2\pi} \epsilon^{i0j} F_{0j} = -\frac{k}{2\pi} \epsilon_{ij} E_j$$

Therefore, the Hall conductivity is,

$$\sigma_{xy} = \frac{k}{2\pi} = \frac{e^2}{2\pi\hbar} \nu \quad (1.201)$$

which is the familiar result! We can also find the charge density,

$$J_0 = \frac{k}{2\pi} B = \frac{e^2 \nu}{2\pi\hbar} B$$

which is also the familiar result!